$$
\begin{aligned}
C_{\phi G}^{(22)} \to \frac{1}{16\,\pi^2} & \Big(\frac{1}{24} \textstyle\sum_p g_1{}^2\, g_3{}^2\, LF_{3,0}\left[m_{\tilde{d}}{}^p\right] - \frac{1}{24} \sum_p g_1{}^2\, g_3{}^2\, LF_{4,-1}\left[m_{\tilde{d}}{}^p\right] - \frac{1}{4}\, g_3{}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\, LF_{3,0}\left[m_{\tilde{d}}{}^r\right] + \\
& \frac{1}{4}\, g_3{}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\, LF_{4,-1}\left[m_{\tilde{d}}{}^r\right] + \frac{1}{24}\, g_3{}^2\, \left(-6\, \overline{y_d}^{pr}\, y_d{}^{pr} + \textstyle\sum_p g_1{}^2\right)\, LF_{3,0}\left[m_{\tilde{q}}{}^p\right] + \\
& \frac{1}{24}\, g_3{}^2\, \left(6\, \overline{y_d}^{pr}\, y_d{}^{pr} - \textstyle\sum_p g_1{}^2\right)\, LF_{4,-1}\left[m_{\tilde{q}}{}^p\right] - \frac{1}{12} \sum_p g_1{}^2\, g_3{}^2\, LF_{3,0}\left[m_{\tilde{u}}{}^p\right] + \frac{1}{12} \sum_p g_1{}^2\, g_3{}^2\, LF_{4,-1}\left[m_{\tilde{u}}{}^p\right] - \\
& \frac{1}{4}\, g_3{}^2\, \overline{a_d}^{pr}\, a_d{}^{pr}\, LF_{2,2,0}\left[m_{\tilde{d}}{}^r,\, m_{\tilde{q}}{}^p\right] - \frac{1}{4}\, g_3{}^2\, \overline{a_d}^{pr}\, a_d{}^{pr}\, LF_{3,1,0}\left[m_{\tilde{d}}{}^r,\, m_{\tilde{q}}{}^p\right] + \\
& \frac{1}{4}\, g_3{}^2\, \overline{a_d}^{pr}\, a_d{}^{pr}\, LF_{3,2,-1}\left[m_{\tilde{d}}{}^r,\, m_{\tilde{q}}{}^p\right] + \frac{1}{4}\, g_3{}^2\, \overline{a_d}^{pr}\, a_d{}^{pr}\, LF_{4,1,-1}\left[m_{\tilde{d}}{}^r,\, m_{\tilde{q}}{}^p\right] - \\
& \frac{1}{4}\, g_3{}^2\, \overline{a_d}^{pr}\, a_d{}^{pr}\, LF_{3,1,0}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{d}}{}^r\right] + \frac{1}{4}\, g_3{}^2\, \overline{a_d}^{pr}\, a_d{}^{pr}\, LF_{3,2,-1}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{d}}{}^r\right] + \\
& \frac{1}{4}\, g_3{}^2\, \overline{a_d}^{pr}\, a_d{}^{pr}\, LF_{4,1,-1}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{d}}{}^r\right] - \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{2,2,0}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{u}}{}^r\right] - \\
& \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{3,1,0}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{u}}{}^r\right] + \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{3,2,-1}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{u}}{}^r\right] + \\
& \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{4,1,-1}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{u}}{}^r\right] - \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{3,1,0}\left[m_{\tilde{u}}{}^r,\, m_{\tilde{q}}{}^p\right] + \\
& \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{3,2,-1}\left[m_{\tilde{u}}{}^r,\, m_{\tilde{q}}{}^p\right] + \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{4,1,-1}\left[m_{\tilde{u}}{}^r,\, m_{\tilde{q}}{}^p\right]\Big)
\end{aligned}
$$